

Existence of infinitely many minimizing n -harmonic maps in homotopy classes via critical exponents

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The 3rd of September 2026

Weak p -harmonic maps

Let $\mathcal{N}$ be a closed Riemannian manifold. For $n \in \mathbb{N}$, a (weak) n -harmonic map $\mathbb{S}^n \rightarrow \mathcal{N}$ is a critical point of

$$E_n(u) = \int_{\mathbb{S}^n} |\nabla u|^n.$$

They are solutions to the corresponding Euler–Lagrange equation

$$-\operatorname{div}(|\nabla u|^{n-2} \nabla u) = |\nabla u|^{n-2} A(u)[\nabla u, \nabla u],$$

where A is the second fundamental form of $\mathcal{N}$.

How to find n -harmonic maps?

Idea: Solve

$$\min_{u \in W^{1,n}(\mathbb{S}^n; \mathcal{N})} E_n(u).$$

Issue: The problem is trivial, minimisers are constants.

New idea: Fix $\alpha \in \pi_n(\mathcal{N})$, and solve

$$\min_{\substack{u \in W^{1,n}(\mathbb{S}^n; \mathcal{N}) \\ u \in \alpha}} E_n(u).$$

Note: homotopy classes are well defined because $n \geq n$. (See Brezis and Nirenberg (1995).)

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Model case: Mappings $\mathbb{S}^n \rightarrow \mathbb{S}^n$

For $d \notin \mathbb{Z}_*$, solve

$$\min_{\substack{u \in W^{1,n}(\mathbb{S}^n; \mathbb{S}^n) \\ \deg u = d}} E_n(u).$$

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Some old results

There exists a minimising harmonic map $\mathbb{S}^2 \rightarrow \mathbb{S}^2$ of degree d for every $d \in \mathbb{Z}$.

If $n \geq 3$, there exists a minimising n -harmonic map $\mathbb{S}^n \rightarrow \mathbb{S}^n$ of degree d if and only if $d \in \{-1, 0, 1\}$.

Reason: Harmonic maps $\mathbb{S}^n \rightarrow \mathbb{S}^n$ have to be conformal, Liouville's theorem (see Lemaire (1975), Xu and Yang (1993)).

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A tool to produce minimisers: The energy identity

We denote $\mathcal{E}_\alpha^n = \inf_{\substack{u \in W^{1,n}(\mathbb{S}^n; \mathcal{N}) \\ u \in \alpha}} E_n(u).$

Theorem

Assume that $\mathcal{N}$ is simply connected, and let $\alpha \in \pi_n(\mathbb{S}^n)$. There exist $\alpha_1, \dots, \alpha_N \in \pi_n(\mathcal{N})$ such that

- 1 $\alpha = \alpha_1 + \dots + \alpha_N;$
- 2 $\mathcal{E}_\alpha^n = \mathcal{E}_{\alpha_1}^n + \dots + \mathcal{E}_{\alpha_N}^n;$
- 3 the α_i admit minimising harmonic maps.

The energy identity is originally due to Sacks and Uhlenbeck (1981). See also Kawai, Nakauchi, Takeuchi (1999).

Infinitely many homotopy classes with minimisers for the Hopf invariant

Theorem (Rivière (1998))

There exist infinitely many homotopy classes of $\pi_3(\mathbb{S}^2)$ having a minimising harmonic map.

Recall: $\pi_3(\mathbb{S}^2) = \mathbb{Z}$, described by the Hopf invariant.

Ingredients:

- The energy identity;
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A general principle

We let $\deg: \pi_n(\mathcal{N}) \rightarrow \mathbb{R}$ be any element of $\text{Hom}(\pi_n(\mathcal{N}); \mathbb{R})$.

Theorem (D., Grzela, Mazowiecka)

Assume that there exists a sequence $(d_k)_{k \in \mathbb{N}}$ such that

$$\lim_{k \rightarrow +\infty} \frac{1}{d_k} \inf_{\substack{u \in W^{1,n}(\mathbb{S}^n; \mathcal{N}) \\ \deg u = d_k}} E_n(u) = 0.$$

Then, there exists infinitely many homotopy classes of $\pi_n(\mathcal{N})$ that admit a minimising harmonic map.

Critical and optimal exponents

For every $\deg \in \text{Hom}(\pi_n(\mathcal{N}); \mathbb{R})$, there exists $p \in (0, 1]$, called the *critical exponent*, such that

$$\deg(u)^p \lesssim E_n(u) \quad \text{for every } u \in W^{1,n}(\mathbb{S}^n; \mathcal{N}).$$

This is connected to rational homotopy theory, Sullivan's minimal models, and the work of Novikov (see also Hardt and Rivière (2008)).

The critical exponent p is *optimal* if there exists a sequence $d_k \rightarrow +\infty$ such that

$$\lim_{k \rightarrow +\infty} \frac{\inf_{\deg u = d_k} E_n(u)}{\log d_k} = p.$$

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Optimality for the Hopf invariant

Proposition (D., Grzela, Mazowiecka)

The critical exponent $\frac{4n-1}{4n}$ for the Hopf invariant on $\mathbb{S}^{2n}$ is optimal.

Corollary

There exist infinitely many homotopy classes of $\pi_{4n-1}(\mathbb{S}^{2n})$ having a minimising harmonic map.

Extra difficulty when $n \geq 2$: Handle the presence of torsion.

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The critical exponent $\frac{2n+1}{2n+2}$ for $\pi_{2n+1}(\mathbb{CP}^n) \simeq \mathbb{Z}$ is optimal.

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There exist infinitely many homotopy classes of $\pi_{2n+1}(\mathbb{CP}^n)$ having a minimising harmonic map.

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Explicit examples of Hopf classes with minimisers

The theorem by Rivière does not allow to determine *which classes* have minimisers.

Recently, Wang and Zhang proved that the Hopf map is a minimiser among maps with Hopf degree equal to 1.

Theorem (Wang and Zhang (2026))

For every $d \in \mathbb{Z}$,

$$\inf_{\deg_H u = d} E_3(u) \geq 8\sqrt{2}\tau^2|d|^{\frac{3}{4}}.$$

Equality is attained for $d = 1$ exactly if $u = H$ is the Hopf map, up to conformal transformation.

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Minimisers for higher degrees

Proposition (D., Grzela, Mazowiecka)

Let $d \geq 2$. If

$$\inf_{\deg_H u = d} E_3(u) < 8\sqrt{2}\tau^2(1 + (d-1)^{\frac{3}{4}}),$$

then there exists a minimiser of degree d .

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Thank you for your attention!